

S1 Answer Key

Step-by-step solutions to all 30 problems.

Algebra

Problem 1

Solve for x : $\frac{2x - 3}{x + 1} = \frac{x + 5}{x - 2}$

Cross-multiply:

$$(2x - 3)(x - 2) = (x + 5)(x + 1)$$

Expand both sides:

$$2x^2 - 4x - 3x + 6 = x^2 + x + 5x + 5$$

$$2x^2 - 7x + 6 = x^2 + 6x + 5$$

Collect terms:

$$x^2 - 13x + 1 = 0$$

Apply the quadratic formula:

$$x = \frac{13 \pm \sqrt{169 - 4}}{2} = \frac{13 \pm \sqrt{165}}{2}$$

Check that neither solution makes a denominator zero ($x \neq -1$, $x \neq 2$). Both solutions are approximately $x \approx 12.92$ and $x \approx 0.077$ — neither is excluded.

$$x = \frac{13 \pm \sqrt{165}}{2}$$

Problem 2

Find all real solutions: $x^4 - 5x^2 + 4 = 0$

Substitute $u = x^2$:

$$u^2 - 5u + 4 = 0$$

Factor:

$$(u - 1)(u - 4) = 0 \implies u = 1 \text{ or } u = 4$$

Back-substitute:

- $x^2 = 1 \implies x = \pm 1$
- $x^2 = 4 \implies x = \pm 2$

$x = -2, -1, 1, 2$

Problem 3

Solve: $\log_2(x + 3) + \log_2(x - 1) = 5$

Combine using the product rule for logarithms:

$$\log_2[(x + 3)(x - 1)] = 5$$

Rewrite in exponential form:

$$(x + 3)(x - 1) = 2^5 = 32$$

Expand and solve:

$$x^2 + 2x - 3 = 32 \implies x^2 + 2x - 35 = 0$$

$$(x + 7)(x - 5) = 0 \implies x = -7 \text{ or } x = 5$$

Check domain — $\log_2$ requires both arguments positive:

- $x = -7$: $x + 3 = -4 < 0$ — **rejected**
- $x = 5$: $x + 3 = 8 > 0$, $x - 1 = 4 > 0$ — **valid**

$$\boxed{x = 5}$$

Problem 4

Partial fraction decomposition of $\frac{5x - 1}{(x - 1)(x + 2)}$

Set up the form:

$$\frac{5x - 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$$

Multiply both sides by $(x - 1)(x + 2)$:

$$5x - 1 = A(x + 2) + B(x - 1)$$

Solve for A — let $x = 1$:

$$5(1) - 1 = A(3) \implies 4 = 3A \implies A = \frac{4}{3}$$

Solve for B — let $x = -2$:

$$5(-2) - 1 = B(-3) \implies -11 = -3B \implies B = \frac{11}{3}$$

$$\boxed{\frac{5x - 1}{(x - 1)(x + 2)} = \frac{4/3}{x - 1} + \frac{11/3}{x + 2}}$$

Limits

Problem 5

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$

Rewrite to apply the standard limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$:

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = \lim_{x \rightarrow 0} 3 \cdot \frac{\sin(3x)}{3x} = 3 \cdot 1$$

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Problem 6

Evaluate $\lim_{x \rightarrow \infty} \frac{4x^3 - 2x + 1}{7x^3 + 5x^2}$

Divide numerator and denominator by x^3 (highest power):

$$\lim_{x \rightarrow \infty} \frac{4 - \frac{2}{x^2} + \frac{1}{x^3}}{7 + \frac{5}{x}} = \frac{4 - 0 + 0}{7 + 0}$$

$$\boxed{\frac{4}{7}}$$

Problem 7

Evaluate $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$

This is $\frac{0}{0}$ form — apply **L'Hôpital's Rule** twice.

First application:

$$\lim_{x \rightarrow 0} \frac{2e^{2x} - 2}{2x} \quad \left(\frac{0}{0} \right)$$

Second application:

$$\lim_{x \rightarrow 0} \frac{4e^{2x}}{2} = \frac{4 \cdot 1}{2}$$

$$\boxed{2}$$

Basic Derivatives

Problem 8

Differentiate $f(x) = 4x^5 - \frac{3}{x^2} + \sqrt{x} - 7$

Rewrite with exponents:

$$f(x) = 4x^5 - 3x^{-2} + x^{1/2} - 7$$

Apply the power rule to each term:

$$f'(x) = 20x^4 - 3(-2)x^{-3} + \frac{1}{2}x^{-1/2} - 0$$

$$f'(x) = 20x^4 + \frac{6}{x^3} + \frac{1}{2\sqrt{x}}$$

Problem 9

Differentiate $f(x) = 3 \sin(x) - 5 \cos(x) + 2 \tan(x)$

Apply standard trig derivative rules:

$$f'(x) = 3 \cos(x) + 5 \sin(x) + 2 \sec^2(x)$$

Problem 10

Differentiate $f(x) = e^{3x} - \ln(x^2 + 1)$

First term — chain rule:

$$\frac{d}{dx} e^{3x} = 3e^{3x}$$

Second term — chain rule:

$$\frac{d}{dx} \ln(x^2 + 1) = \frac{2x}{x^2 + 1}$$

$$f'(x) = 3e^{3x} - \frac{2x}{x^2 + 1}$$

Chain, Product, and Quotient Rules

Problem 11

Find $\frac{dy}{dx}$ for $y = x^3 \ln(x) - e^{2x} \sin(x)$

First term — product rule:

$$\frac{d}{dx} [x^3 \ln(x)] = 3x^2 \ln(x) + x^3 \cdot \frac{1}{x} = 3x^2 \ln(x) + x^2$$

Second term — product rule with chain rule on e^{2x} :

$$\frac{d}{dx} [e^{2x} \sin(x)] = 2e^{2x} \sin(x) + e^{2x} \cos(x)$$

Combine:

$$\frac{dy}{dx} = 3x^2 \ln(x) + x^2 - e^{2x} (2 \sin(x) + \cos(x))$$

Problem 12

Find $\frac{dy}{dx}$ for $y = \frac{x^2 - 1}{\sin(x)}$

Apply the quotient rule $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$:

$$\frac{dy}{dx} = \frac{2x \cdot \sin(x) - (x^2 - 1) \cos(x)}{\sin^2(x)}$$

$\frac{dy}{dx} = \frac{2x \sin(x) - (x^2 - 1) \cos(x)}{\sin^2(x)}$
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Problem 13

Find $\frac{dy}{dx}$ for $y = \cos^3(4x^2 + 1)$

Three nested functions — apply the chain rule twice.

Let $u = \cos(4x^2 + 1)$, so $y = u^3$:

$$\frac{dy}{dx} = 3u^2 \cdot \frac{du}{dx}$$

Now $\frac{du}{dx} = -\sin(4x^2 + 1) \cdot 8x$:

$$\frac{dy}{dx} = 3 \cos^2(4x^2 + 1) \cdot (-\sin(4x^2 + 1)) \cdot 8x$$

$$\boxed{\frac{dy}{dx} = -24x \cos^2(4x^2 + 1) \sin(4x^2 + 1)}$$

Implicit Differentiation

Problem 14

Find $\frac{dy}{dx}$ for $x^2y + y^3 = 6$

Differentiate both sides with respect to x :

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

Collect $\frac{dy}{dx}$ terms:

$$\frac{dy}{dx}(x^2 + 3y^2) = -2xy$$

$$\boxed{\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}}$$

Problem 15

For $x^3 + y^3 = 3xy$, find $\frac{dy}{dx}$ and the tangent line at $\left(\frac{3}{2}, \frac{3}{2}\right)$.

(Verify the point: $\frac{27}{8} + \frac{27}{8} = \frac{27}{4}$ and $3 \cdot \frac{3}{2} \cdot \frac{3}{2} = \frac{27}{4}$ ✓)

Differentiate both sides:

$$3x^2 + 3y^2 \frac{dy}{dx} = 3y + 3x \frac{dy}{dx}$$

Isolate $\frac{dy}{dx}$:

$$3y^2 \frac{dy}{dx} - 3x \frac{dy}{dx} = 3y - 3x^2 \implies \frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

Evaluate at $\left(\frac{3}{2}, \frac{3}{2}\right)$:

$$\frac{dy}{dx} = \frac{\frac{3}{2} - \frac{9}{4}}{\frac{9}{4} - \frac{3}{2}} = \frac{-\frac{3}{4}}{\frac{3}{4}} = -1$$

Tangent line:

$$y - \frac{3}{2} = -1 \left(x - \frac{3}{2} \right) \implies y = -x + 3$$

$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}, \quad \text{tangent line: } y = -x + 3$

Applied Optimization

Problem 16

Find the absolute max and min of $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 4]$.

Find critical points — set $f'(x) = 0$:

$$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$$

Critical points: $x = 3$ and $x = -1$ — both in $[-2, 4]$.

Evaluate f at critical points and endpoints:

$$f(-2) = -8 - 12 + 18 + 5 = 3$$

$$f(-1) = -1 - 3 + 9 + 5 = 10$$

$$f(3) = 27 - 27 - 27 + 5 = -22$$

$$f(4) = 64 - 48 - 36 + 5 = -15$$

Absolute maximum: $f(-1) = 10$	Absolute minimum: $f(3) = -22$
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Problem 17

$$s(t) = t^3 - 6t^2 + 9t, \quad t \geq 0.$$

Velocity:

$$v(t) = s'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

At rest: $v(t) = 0 \implies t = 1$ and $t = 3$

Moving in positive direction: $v(t) > 0$ when $t \in [0, 1)$ or $t \in (3, \infty)$

Total distance on $[0, 4]$ — evaluate s at each rest time and endpoint:

$$s(0) = 0, \quad s(1) = 1 - 6 + 9 = 4, \quad s(3) = 27 - 54 + 27 = 0, \quad s(4) = 64 - 96 + 36 = 4$$

$$d = |s(1) - s(0)| + |s(3) - s(1)| + |s(4) - s(3)| = 4 + 4 + 4$$

At rest at $t = 1, t = 3$ Total distance = 12 units

Basic Integrals

Problem 18

Evaluate $\int \left(6x^2 - \frac{4}{x} + 3e^x \right) dx$

Integrate term by term:

$$\int 6x^2 dx = 2x^3, \quad \int \frac{4}{x} dx = 4 \ln |x|, \quad \int 3e^x dx = 3e^x$$

$2x^3 - 4 \ln x + 3e^x + C$

Problem 19

Evaluate $\int_0^\pi \sin(x) \, dx$

$$\int_0^\pi \sin(x) \, dx = \left[-\cos(x) \right]_0^\pi = -\cos(\pi) + \cos(0) = 1 + 1$$

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Problem 20

Find $F'(x)$ where $F(x) = \int_1^{x^2} \frac{t}{t^2 + 1} dt$

Apply the Fundamental Theorem of Calculus with the chain rule. Let $g(x) = x^2$:

$$F'(x) = \frac{g(x)}{[g(x)]^2 + 1} \cdot g'(x) = \frac{x^2}{x^4 + 1} \cdot 2x$$

$$F'(x) = \frac{2x^3}{x^4 + 1}$$

U-Substitution

Problem 21

Evaluate $\int \frac{3x^2 + 2}{x^3 + 2x + 1} dx$

Let $u = x^3 + 2x + 1$, then $du = (3x^2 + 2) dx$:

$$\int \frac{du}{u} = \ln |u| + C$$

$$\boxed{\ln |x^3 + 2x + 1| + C}$$

Problem 22

Evaluate $\int_0^1 x e^{x^2} dx$

Let $u = x^2$, then $du = 2x dx$, so $x dx = \frac{du}{2}$.

Change limits: $x = 0 \rightarrow u = 0$; $x = 1 \rightarrow u = 1$

$$\int_0^1 x e^{x^2} dx = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2} \left[e^u \right]_0^1 = \frac{1}{2} (e - 1)$$

$$\boxed{\frac{e - 1}{2}}$$

Problem 23

Evaluate $\int \sin^4(x) \cos(x) \, dx$

Let $u = \sin(x)$, then $du = \cos(x) \, dx$:

$$\int u^4 \, du = \frac{u^5}{5} + C$$

$$\boxed{\frac{\sin^5(x)}{5} + C}$$

Integration by Parts

Problem 24

Evaluate $\int_0^{\pi/2} x \cos(x) dx$

Let $u = x, dv = \cos(x) dx \Rightarrow du = dx, v = \sin(x)$

$$\begin{aligned}\int_0^{\pi/2} x \cos(x) dx &= \left[x \sin(x) \right]_0^{\pi/2} - \int_0^{\pi/2} \sin(x) dx \\&= \frac{\pi}{2} \cdot 1 - 0 - \left[-\cos(x) \right]_0^{\pi/2} \\&= \frac{\pi}{2} - \left(-\cos\left(\frac{\pi}{2}\right) + \cos(0) \right) = \frac{\pi}{2} - (0 + 1)\end{aligned}$$

$$\boxed{\frac{\pi}{2} - 1}$$

Problem 25

Evaluate $\int x^2 e^x dx$

First pass — let $u = x^2$, $dv = e^x dx$:

$$\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx$$

Second pass on $\int 2x e^x dx$ — let $u = 2x$, $dv = e^x dx$:

$$\int 2x e^x dx = 2x e^x - \int 2e^x dx = 2x e^x - 2e^x$$

Combine:

$$\int x^2 e^x dx = x^2 e^x - 2x e^x + 2e^x + C$$

$$\boxed{e^x(x^2 - 2x + 2) + C}$$

Differential Equations

Problem 26

Solve $\frac{dy}{dx} = \frac{x^2}{y}$, $y(0) = 3$

Separate variables:

$$y \, dy = x^2 \, dx$$

Integrate both sides:

$$\frac{y^2}{2} = \frac{x^3}{3} + C$$

Apply initial condition $y(0) = 3$:

$$\frac{9}{2} = 0 + C \implies C = \frac{9}{2}$$

Solve for y :

$$y^2 = \frac{2x^3}{3} + 9$$

$$y = \sqrt{\frac{2x^3}{3} + 9}$$

Problem 27

Solve $\frac{dy}{dx} + 2y = 4e^{-x}$

Identify integrating factor $\mu = e^{\int 2 dx} = e^{2x}$

Multiply both sides by μ :

$$\frac{d}{dx} [e^{2x} y] = 4e^{-x} \cdot e^{2x} = 4e^x$$

Integrate both sides:

$$e^{2x} y = 4e^x + C$$

Divide by e^{2x} :

$$y = 4e^{-x} + Ce^{-2x}$$

Problem 28

Find the general solution of $y'' - 5y' + 6y = 0$

Write the characteristic equation:

$$r^2 - 5r + 6 = 0 \implies (r - 2)(r - 3) = 0$$

Distinct real roots: $r_1 = 2, r_2 = 3$

$$y = C_1 e^{2x} + C_2 e^{3x}$$

Problem 29

Solve $y'' + 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = -1$

Characteristic equation:

$$r^2 + 4r + 4 = 0 \implies (r + 2)^2 = 0$$

Repeated root $r = -2$, so the general solution is:

$$y = (C_1 + C_2x)e^{-2x}$$

Apply $y(0) = 1$:

$$C_1 = 1$$

Differentiate:

$$y' = C_2e^{-2x} - 2(C_1 + C_2x)e^{-2x}$$

Apply $y'(0) = -1$:

$$C_2 - 2C_1 = -1 \implies C_2 - 2 = -1 \implies C_2 = 1$$

$$\boxed{y = (1 + x)e^{-2x}}$$

Problem 30

Salt tank: $Q(0) = 10$ kg, volume = 100 L, flow in/out = 5 L/min (pure water in).

Set up the IVP. The rate of salt leaving is:

$$\begin{aligned}\text{rate out} &= 5 \frac{\text{L}}{\text{min}} \cdot \frac{Q}{100} \frac{\text{kg}}{\text{L}} = \frac{Q}{20} \frac{\text{kg}}{\text{min}} \\ \frac{dQ}{dt} &= -\frac{Q}{20}, \quad Q(0) = 10\end{aligned}$$

Separate and integrate:

$$\frac{dQ}{Q} = -\frac{dt}{20} \implies \ln|Q| = -\frac{t}{20} + C$$

General solution:

$$Q(t) = Ae^{-t/20}$$

Apply $Q(0) = 10$: $A = 10$

$$Q(t) = 10e^{-t/20}$$

Find t when $Q = 1$ kg:

$$1 = 10e^{-t/20} \implies e^{-t/20} = \frac{1}{10} \implies t = 20 \ln(10)$$

$Q(t) = 10e^{-t/20} \text{ kg}, \quad t = 20 \ln(10) \approx 46.1 \text{ minutes}$
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